

CIE-1 PYQ Bank: Model Answers

Handwritten Question Bank & Solutions

Syllabus Topics Covered:

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Unit 1 Examination Questions & Answers

UNIT 1

Verified model solutions for Unit 1 past exam questions.

Unit 1 6 Marks High Priority

Ramaiah CIE-1 2026 Q1.a (6M)SEE June/July 2026 Q2.a (7M)SEE May 2023 Q1 (6M)

[Theory Link: Unit 1 § 1](#)

Explain the difference between Supervised and Unsupervised Learning with suitable examples. Compare them with Reinforcement Learning.

Practice Problem:

Model Exam Answer

Solution:**Definitions:**

Supervised: Supervised Learning: The algorithm is provided with a labeled dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, where each input vector $\mathbf{x}_i$ is paired with a known ground-truth target label y_i . The goal is to learn a mapping function $f: \mathcal{X} \rightarrow \mathcal{Y}$ that generalizes accurately to unseen test instances.

Unsupervised: Unsupervised Learning: The algorithm is provided with unlabeled data $\mathcal{D} = \{\mathbf{x}_i\}_{i=1}^N$ without target annotations. The objective is to discover underlying patterns, intrinsic structures, clusters, or probability distributions.

Reinforcement: Reinforcement Learning: An autonomous agent interacts with an environment through trial-and-error actions a_t , observing state transitions s_{t+1} and scalar reward signals r_t . The goal is to learn an optimal policy $\pi^*(s)$ that maximizes cumulative expected long-term discounted reward.

Comparison Table:**Headers:**

- Attribute
- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

Rows:

- ['Training Data', 'Input data paired with explicit ground-truth labels ($\mathbf{x}, y$)', 'Input data only $\mathbf{x}$; no labels or supervision.', 'No fixed dataset; dynamic state-action-reward']

Unit 1 6 Marks High Priority

SEE June/July 2026 Q1.a (6M) SEE July 2024 Q1 (6M)

Theory Link: Unit 1 § 1

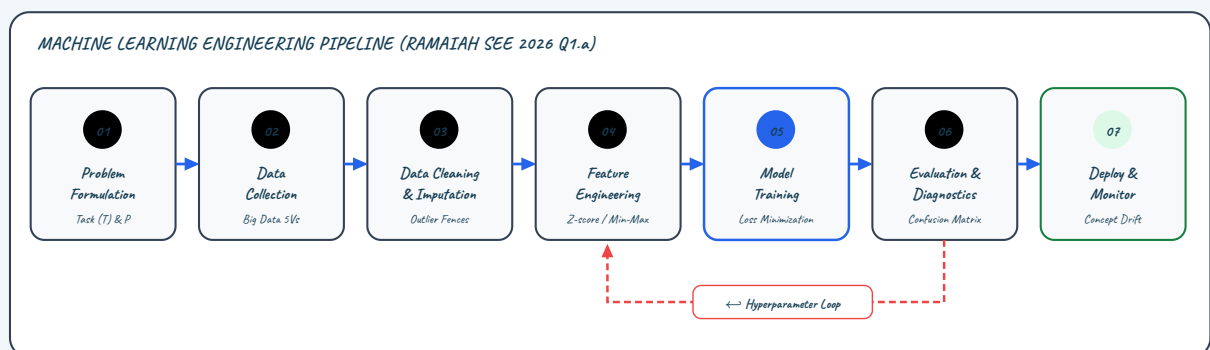
Describe the complete 7-step Machine Learning process pipeline with a neat diagram.

Practice Problem:

Model Exam Answer

Solution:

The machine learning process is a disciplined engineering workflow transforming raw domain data into a reliable, deployed predictive model.

Seven Steps:

- **1. Data Collection:** Gather raw data from diverse sources (databases, APIs, IoT sensors, web scraping).
- **2. Data Preparation & Cleaning:** Handle missing values, remove noise/outliers, normalize features, split data into Train (70%), Validation (15%), and Test (15%) subsets.
- **3. Model Selection:** Choose suitable candidate model families (e.g., Linear Regression, Logistic Regression, Random Forest) based on data size, type, and complexity.
- **4. Model Training:** Optimize model parameters $\mathbf{w}$ using training data by minimizing an objective loss function via Gradient Descent or closed-form OLS.
- **5. Model Evaluation:** Benchmark the model against unseen validation data using objective metrics (MSE, R^2 , Accuracy, F1-Score, ROC-AUC).
- **6. Hyperparameter Tuning:** Optimize non-trainable hyperparameters (learning rate α , regularization strength λ) using grid search or cross-validation.
- **7. Deployment & Monitoring:** Deploy model as a REST API or microservice;

Unit 1 8 Marks High Priority

Ramaiah CIE-1 2026 Q2.a (6M)SEE June/July 2026 Q1.b (8M)

Theory Link: Unit 1 § 1

A binary classifier produces the following confusion matrix: $TP = 70$, $TN = 50$, $FP = 20$, $FN = 10$. Compute Accuracy, Precision, Recall, F1-Score, Sensitivity, and Specificity. Discuss suitability for medical diagnosis.

Practice Problem:

Model Exam Answer

Solution:

Given:

$TP = 70$, $TN = 50$, $FP = 20$, $FN = 10$. Total instances $N = 70 + 50 + 20 + 10 = 150$.

Computations:

- **1. Accuracy:** Proportion of total correct predictions.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{70 + 50}{150} = \frac{120}{150} = \mathbf{0.80}$$

- **2. Precision:** Proportion of predicted positives that are actually positive.

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{70}{70 + 20} = \frac{70}{90} = \mathbf{0.7778 \text{ (77.78\%)}}$$

- **3. Recall / Sensitivity:** Proportion of actual positives identified.

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{70}{70 + 10} = \frac{70}{80} = \mathbf{0.8750 \text{ (87.50\%)}}$$

- **4. Specificity (True Negative Rate):** Proportion of actual negatives correctly identified.

$$\text{Specificity} = \frac{TN}{TN + FP} = \frac{50}{50 + 20} = \frac{50}{70} = \mathbf{0.7143 \text{ (71.43\%)}}$$

- **5. F1-Score:** Harmonic mean of Precision and Recall.

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \times 0.7778 \times 0.8750}{0.7778 + 0.8750} = \frac{1.3611}{1.6528}$$

Medical Suitability Commentary:

Critical Application Analysis: In life-threatening medical diagnosis (e.g., malignant tumor detection), **False Negatives ($FN = 10$)** are disastrous because sick patients are sent home untreated. While Recall is respectable at 87.5%, an FN rate of 12.5% is

Unit 1 6 Marks High Priority

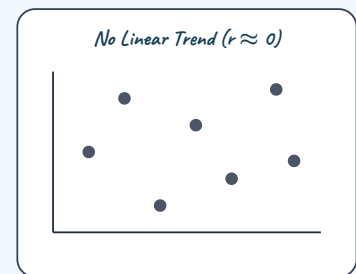
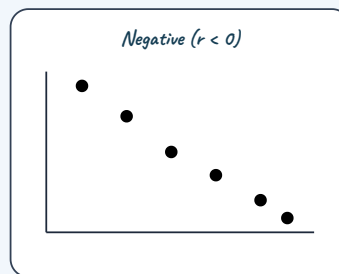
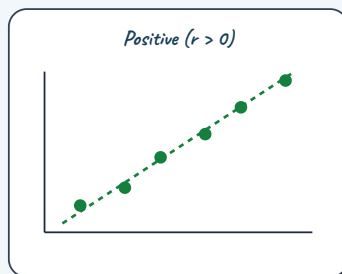
Shruti mam Docx Problem 1 & 3 Unit-1 QB (6M)

Theory Link: Unit 1 § 1

Define Covariance and Pearson's Correlation Coefficient. The following data show hours studied (X) and marks obtained (Y) by four students: $X = [2, 4, 6, 8]$, $Y = [50, 60, 70, 80]$. Calculate Covariance and Pearson's r with step-by-step verification.

Practice Problem:

Model Exam Answer

Solution:Definitions:

- **Sample Covariance:** Measures the directional joint variability of two random variables:

$$\text{Cov}(X, Y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$

- **Pearson's Correlation Coefficient (r):** Scale-invariant measure of linear association normalized between $[-1, +1]$:

$$r = \frac{\text{Cov}(X, Y)}{s_X s_Y} = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2 \sum (Y_i - \bar{Y})^2}}$$

Given Data:

$n = 4$. $X = [2, 4, 6, 8]$, $Y = [50, 60, 70, 80]$.

Step By Step:

- 1. **Compute Means:**

$$\bar{X} = \frac{2 + 4 + 6 + 8}{4} = 5.0, \quad \bar{Y} = \frac{50 + 60 + 70 + 80}{4} = 65.0$$

Unit 1 6 Marks Medium Priority

Sridhar & Vijayalakshmi Ch. 2.4 Unit-1 QB (6M)

[Theory Link: Unit 1 § 1](#)

Explain the Five-Number Summary and Outlier Detection using the $1.5 \times \text{IQR}$ rule with Box Plots.

Practice Problem:

Model Exam Answer

Solution:*Five Number Summary:*

- **1. Minimum:** Lowest data value excluding outliers ($Q_1 - 1.5 \times \text{IQR}$).
- **2. First Quartile (Q_1 / 25th Percentile):** Median of the lower half of sorted data.
- **3. Median (Q_2 / 50th Percentile):** Middle value dividing data into two equal halves.
- **4. Third Quartile (Q_3 / 75th Percentile):** Median of the upper half of sorted data.
- **5. Maximum:** Highest data value excluding outliers ($Q_3 + 1.5 \times \text{IQR}$).

Iqr Outlier Rule:

- **Interquartile Range:** $\text{IQR} = Q_3 - Q_1$, representing the spread of the middle 50% of observations.
- **Inner Fences for Outliers:**
 - Lower Fence = $Q_1 - 1.5 \times \text{IQR}$
 - Upper Fence = $Q_3 + 1.5 \times \text{IQR}$
- Any data point $x < \text{Lower Fence}$ or $x > \text{Upper Fence}$ is flagged as a statistical outlier and plotted as an individual point beyond the whiskers in a Box Plot.

Source: Sridhar & Vijayalakshmi Ch. 2.4 (pp. 52-56)

Unit 2 Examination Questions & Answers

UNIT 2

Verified model solutions for Unit 2 past exam questions.

Unit 2 6 Marks High Priority

Ramaiah CIE-1 2026 Q1.b (6M)SEE June/July 2026 Q3.a (6M)

Theory Link: Unit 2 § 1

Explain how dissimilarity is computed for Ordinal attributes. Given four objects with test results: Object 1 = Excellent, Object 2 = Fair, Object 3 = Good, Object 4 = Excellent. Rank order: Fair < Good < Excellent. Compute the normalized values and the pairwise dissimilarity matrix using step-by-step formulas.

Practice Problem:

Model Exam Answer

Solution:Formula Explanation:

- **Step 1: Replace with Rank:** For an ordinal attribute f with M_f ordered states, assign each state an integer rank $r_{if} \in \{1, 2, \dots, M_f\}$.
- **Step 2: Map to Normalized Interval $[0, 1]$:** Apply the formula:

$$z_{if} = \frac{r_{if} - 1}{M_f - 1}$$

- **Step 3: Compute Pairwise Dissimilarity:** For single attribute, dissimilarity is the absolute difference:

$$d(i, j) = |z_{if} - z_{jf}|$$

Step By Step Calculation:

- States: $M = 3$ (Fair = 1, Good = 2, Excellent = 3). Range $M - 1 = 2$.
- Normalized Scores z_i :
 - Object 1 (Excellent): $r_1 = 3 \implies z_1 = (3 - 1)/2 = 1.0$
 - Object 2 (Fair): $r_2 = 1 \implies z_2 = (1 - 1)/2 = 0.0$
 - Object 3 (Good): $r_3 = 2 \implies z_3 = (2 - 1)/2 = 0.5$
 - Object 4 (Excellent): $r_4 = 3 \implies z_4 = (3 - 1)/2 = 1.0$
- Pairwise Dissimilarities $d(i, j) = |z_i - z_j|$:
 - $d(1, 1) = 0.0$
 - $d(2, 1) = |0.0 - 1.0| = 1.0$
 - $d(3, 1) = |0.5 - 1.0| = 0.5$, $d(3, 2) = |0.5 - 0.0| = 0.5$
 - $d(4, 1) = |1.0 - 1.0| = 0.0$, $d(4, 2) = |1.0 - 0.0| = 1.0$, $d(4, 3) = |1.0 - 0.5| = 0.5$

Dissimilarity Matrix:

Unit 2 6 Marks High Priority

Ramaiah CIE-2 2026 Q2.c (6M)SEE June/July 2026 Q9.a (7M)

[Theory Link: Unit 2 § 4](#)

Using the covariance matrix $C = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$, apply Principal Component Analysis (PCA): 1. Find the Eigenvalues, 2. Find the normalized Eigenvectors, 3. Identify the First Principal Component (PC1) and calculate the proportion of total variance it explains.

Practice Problem:

Model Exam Answer

Solution:

Given:

Covariance matrix $C = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$.

Step 1 Eigenvalues:

- Solve characteristic equation $\det(C - \lambda I) = 0$:
- $\det \begin{pmatrix} 4 - \lambda & 2 \\ 2 & 4 - \lambda \end{pmatrix} = (4 - \lambda)^2 - (2)(2) = 0$
- $\lambda^2 - 8\lambda + 16 - 4 = 0 \implies \lambda^2 - 8\lambda + 12 = 0$
- $(\lambda - 6)(\lambda - 2) = 0 \implies \lambda_1 = 6, \quad \lambda_2 = 2$

Step 2 Eigenvectors:

- For $\lambda_1 = 6$: $(C - 6I)\mathbf{v}_1 = \mathbf{0}$:

$$\begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} v_{11} \\ v_{12} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies -2v_{11} + 2v_{12} = 0 \implies v_{11} = v_{12}$$

Unnormalized: $\mathbf{v}_1 = [1, 1]^T$. Length = $\sqrt{1^2 + 1^2} = \sqrt{2}$.

Normalized: $\mathbf{u}_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \approx \begin{bmatrix} 0.7071 \\ 0.7071 \end{bmatrix}$.

- For $\lambda_2 = 2$: $(C - 2I)\mathbf{v}_2 = \mathbf{0}$:

$$\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} v_{21} \\ v_{22} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies 2v_{21} + 2v_{22} = 0 \implies v_{21} = -v_{22}$$

Unit 2 10 Marks High Priority

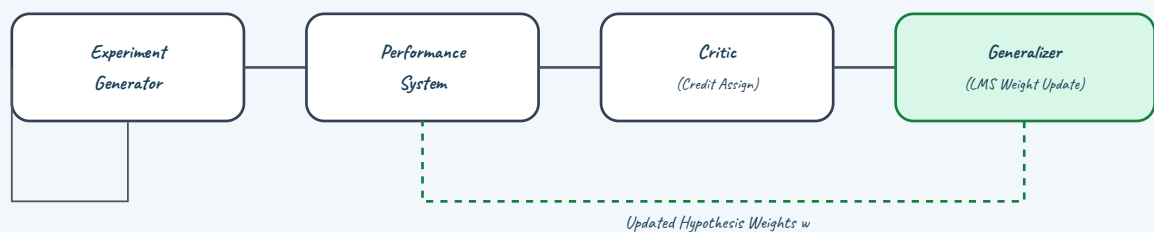
Unit-1 QB Q2.a & Q2.b (10M) SEE May 2023 Q2 (10M)

Theory Link: Unit 2 § 2

Describe the five-step methodology for designing a learning system from Tom Mitchell. Illustrate each step using the Checkers Learning Problem.

Practice Problem:

Model Exam Answer

Solution:Mitchell Definition:

A computer program is said to learn from **Experience E** with respect to some class of **Tasks T** and **Performance measure P**, if its performance at tasks in T, as measured by P, improves with experience E.

Five Steps:

- **1. Choosing the Training Experience:** Select source and feedback type (Direct vs Indirect; Teacher-guided vs Self-play games). In Checkers: Self-play games provide indirect feedback (win/loss signal only at game end).
- **2. Choosing the Target Function:** Formalize what knowledge to learn. In Checkers: Function $V : B \rightarrow \mathbb{R}$ mapping any board state $b \in B$ to a real evaluation score (+100 for win, -100 for loss, intermediate values for board advantage).
- **3. Choosing a Representation for Target Function:** Select mathematical hypothesis representation. In Checkers: Linear combination of board features: $\hat{V}(b) = w_0 + w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + w_5x_5 + w_6x_6$, where x_1 = black pieces, x_2 = red pieces, x_3 = black kings, etc.
- **4. Choosing a Function Approximation Algorithm:** Select learning rule to adjust weights w_i . In Checkers: LMS (Least Mean Squares) weight update rule: $w_i \leftarrow w_i + \eta(V_{\text{train}}(b) - \hat{V}(b))x_i$.

Unit 2 6 Marks High Priority

Ramaiah CIE-1 2026 Q2.b (6M)SEE June/July 2026 Q4.b (6M)

Theory Link: Unit 2 § 3

Explain Concept Learning and the Find-S Algorithm. Consider a 4-attribute dataset for Buys_Computer (Age, Income, Student, Credit_Rating). Initialize the most specific hypothesis and trace Find-S step by step across positive examples to derive the final hypothesis.

Practice Problem:

Model Exam Answer

Solution:Concept Learning Def:

Concept Learning is acquiring the definition of a general category given a sample of positive and negative training examples of the category. It is equivalent to searching through a predefined hypothesis space H for a hypothesis consistent with training examples.

Find S Rules:

- 1. Initialize hypothesis h to the most specific hypothesis in H : $h = \langle \emptyset, \emptyset, \dots, \emptyset \rangle$.
- 2. For each positive training instance x :
 - For each attribute constraint a_i in h :
 - If a_i is satisfied by x , do nothing.
 - Else, replace a_i in h by the next more general constraint that is satisfied by x (e.g. replace specific value with '?' wildcard).
- 3. Negative training instances are completely ignored by Find-S.
- 4. Output final hypothesis h .

Step By Step Trace:

- **Initialization:** $h_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$
- **Instance 1:** $\langle \text{Youth, High, No, Fair} \rangle \rightarrow \text{No (Negative} \implies \text{IGNORE):}$
 $h_1 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$
- **Instance 2:** $\langle \text{Senior, Low, Yes, Fair} \rangle \rightarrow \text{Yes (Positive! First update):}$
 $h_2 = \langle \text{Senior, Low, Yes, Fair} \rangle$
- **Instance 3:** $\langle \text{Middle, Low, Yes, Excellent} \rangle \rightarrow \text{Yes (Positive!):}$
 - Age: Senior vs Middle $\rightarrow$ generalized to '?'
 - Income: Low vs Low $\rightarrow$ remains 'Low'
 - Student: Yes vs Yes $\rightarrow$ remains 'Yes'

Unit 2 8 Marks High Priority

Ramaiah CIE-1 2026 Q3.b (6M) SEE June/July 2026 Q3.b (8M)

[Theory Link: Unit 2 § 3](#)

Explain the Candidate Elimination Algorithm in Concept Learning. Describe how the Specific boundary (S) and General boundary (G) are updated for positive and negative examples.

Practice Problem:

Model Exam Answer

Solution:

The **Candidate-Elimination Algorithm** computes the entire Version Space (set of all hypotheses consistent with training examples) by maintaining two compact boundary sets: the **Specific boundary (S)** and the **General boundary (G)**.

Boundary Definitions:

- **Specific Boundary S :** Set of minimally general (most specific) hypotheses consistent with all observed instances. Initialized to $S_0 = \{\langle \emptyset, \emptyset, \dots \rangle\}$.
- **General Boundary G :** Set of maximally general hypotheses consistent with all observed instances. Initialized to $G_0 = \{\langle ?, ?, \dots \rangle\}$.

Update Rules:

- **For a POSITIVE training example $d = (x, +)$:**
 1. Remove any hypothesis from G that is not consistent with d .
 2. For each hypothesis $s \in S$ not consistent with d : remove s and add all minimal generalizations h of s such that h is consistent with d and some member of G is more general than h .
 3. Remove from S any hypothesis that is more general than another hypothesis in S .
- **For a NEGATIVE training example $d = (x, -)$:**
 1. Remove any hypothesis from S that is consistent with d .
 2. For each hypothesis $g \in G$ that is consistent with d : remove g and add all minimal specializations h of g such that h is not consistent with d and some member of S is more specific than h .
 3. Remove from G any hypothesis that is more specific than another hypothesis in G .

Stopping Condition:

When the algorithm converges ($S = G = \{c\}$), the target concept c is uniquely learned. If S and G become empty, the training data is noisy or inconsistent with hypothesis

Unit 2 7 Marks High Priority

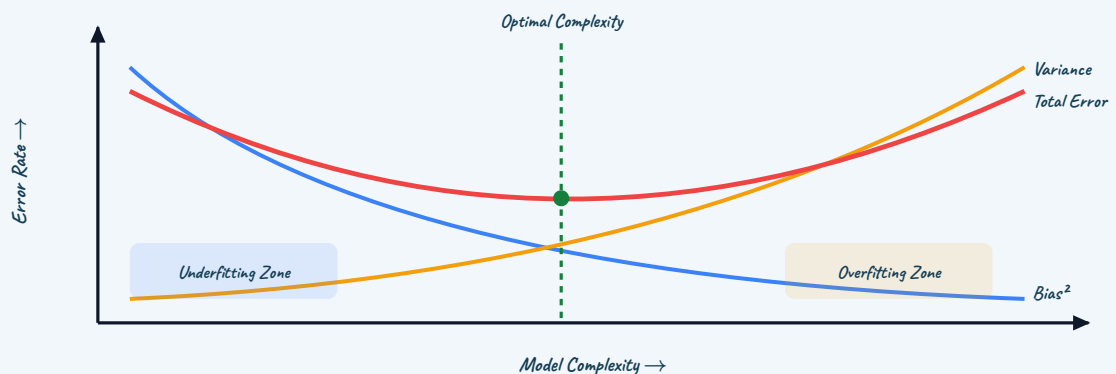
SEE June/July 2026 Q1.c (6M) SEE June/July 2026 Q2.b (7M) Ramaiah CIE-1 2026 Q3.a (6M)

Theory Link: Unit 2 § 4

Explain how Bias and Variance change with model complexity. A model achieves Training Accuracy = 98% and Validation Accuracy = 72%. Diagnose whether this is overfitting or underfitting, suggest two methods to improve it, and explain the roles of train, val, and test datasets.

Practice Problem:

Model Exam Answer

Solution:**Diagnosis: OVERFITTING (High Variance).**

A large performance discrepancy between near-perfect Training Accuracy (98%) and degraded Validation Accuracy (72%) is the classic hallmark of overfitting. The model has memorized idiosyncrasies and random noise in the training set instead of learning true underlying relationships.

Two Remedies:

- **1. Regularization (L2 Ridge / L1 Lasso / Dropout):** Penalize large parameter weights $\lambda \sum w_j^2$, shrinking model complexity and smoothing decision boundaries.
- **2. Early Stopping & Pruning / Feature Reduction:** Stop training before over-optimization, use PCA/Lasso to prune redundant input features, or gather additional diverse training samples.

Dataset Roles:

- **Training Set (70%):** Used by the optimization algorithm to update model weights/coefficients directly.

Unit 3 Examination Questions & Answers

UNIT 3

Verified model solutions for Unit 3 past exam questions.

Unit 3 8 Marks High Priority

Ramaiah CIE-1 2026 Q1.c (3M) Faculty docx Problem 1SEE May 2023 Q1 (8M)

Theory Link: Unit 3 § Linear-regression

The following data represents hours studied (X) and marks obtained (Y) by five students: $X = [2, 4, 6, 8, 10]$, $Y = [50, 60, 65, 80, 85]$. i. Compute slope (b) and intercept (a) using Ordinary Least Squares (OLS). ii. Write the regression equation. iii. Predict marks for $X = 7$ hours. iv. Compute the Coefficient of Determination (R^2).

Practice Problem:

Model Exam Answer

Solution:

Given Data:

Five data points ($n = 5$):

$$X = [2, 4, 6, 8, 10], Y = [50, 60, 65, 80, 85].$$

Step 1 Summations:

- $\sum X = 2 + 4 + 6 + 8 + 10 = 30 \implies \bar{X} = 30/5 = 6.0$
- $\sum Y = 50 + 60 + 65 + 80 + 85 = 340 \implies \bar{Y} = 340/5 = 68.0$
- $\sum X^2 = 2^2 + 4^2 + 6^2 + 8^2 + 10^2 = 4 + 16 + 36 + 64 + 100 = 220$
- $\sum XY = (2)(50) + (4)(60) + (6)(65) + (8)(80) + (10)(85) = 100 + 240 + 390 + 640 + 850 = 2220$

Step 2 Slope And Intercept:

- Slope (b_1 or b):

$$b = \frac{n \sum XY - (\sum X)(\sum Y)}{n \sum X^2 - (\sum X)^2} = \frac{5(2220) - (30)(340)}{5(220) - (30)^2} = \frac{11100}{1100}$$

- Intercept (b_0 or a):

$$a = \bar{Y} - b\bar{X} = 68.0 - (4.5000)(6.0) = 68.0 - 27.0 = 41.0000$$

Step 3 Equation And Prediction:

- Regression Equation:

$$\hat{Y} = 41.0 + 4.5X$$

- Prediction for $X = 7$ hours:

$$\hat{Y}(7) = 41.0 + 4.5(7) = 41.0 + 31.5 = 72.50 \text{ marks}$$

Unit 3 8 Marks High Priority

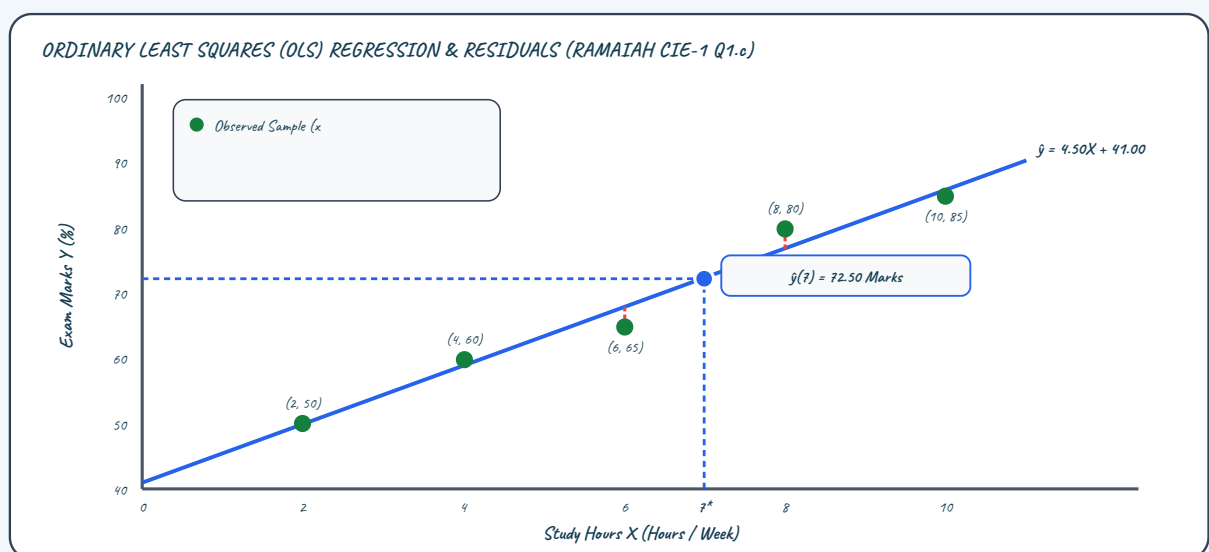
SEE May 2023 Q1 (10M) Faculty PPT Chapter 5.1

Theory Link: Unit 3 § Linear-regression

Derive the closed-form formulas for slope (b_1) and intercept (b_0) in Simple Linear Regression using Ordinary Least Squares (OLS) calculus minimization.

Practice Problem:

Model Exam Answer

Solution:

i, y_i) OLS Fitted Regression Line Residual $e_i = y_i - \hat{y}_i$

Model Equation:

Simple linear regression model: $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$. Fitted value $\hat{y}_i = \beta_0 + \beta_1 x_i$

Objective:

Minimize the Sum of Squared Residuals (Residual Sum of Squares, RSS / SSE):

$$S(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

Derivation Beta0:

- Differentiate S partially with respect to β_0 and set to zero:

$$\frac{\partial S}{\partial \beta_0} = -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0$$

Unit 3 8 Marks High Priority

SEE June/July 2026 Q5.a (8M)Faculty docx Problem 1

Theory Link: Unit 3 § multiple-regression

Formulate the Multiple Linear Regression model in matrix notation. State the closed-form Normal Equations solution. How are partial regression coefficients interpreted?

Practice Problem:

Model Exam Answer

Solution:*Matrix Formulation:*

- For n observations and p predictor variables:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \epsilon_i$$

- In compact matrix notation:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

- where $\mathbf{Y} \in \mathbb{R}^{n \times 1}$, $\mathbf{X} \in \mathbb{R}^{n \times (p+1)}$ (with leading column of ones for intercept), $\boldsymbol{\beta} \in \mathbb{R}^{(p+1) \times 1}$, and error $\boldsymbol{\epsilon} \in \mathbb{R}^{n \times 1}$.

Normal Equations:

- To minimize residual sum of squares $S(\boldsymbol{\beta}) = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})$:
- $\text{del}_{\boldsymbol{\beta}} S = -2\mathbf{X}^T (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) = \mathbf{0} \implies \mathbf{X}^T \mathbf{X} \boldsymbol{\beta} = \mathbf{X}^T \mathbf{Y}$
- Provided $(\mathbf{X}^T \mathbf{X})$ is non-singular (no exact multicollinearity):

$$\boldsymbol{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$$

Partial Slope Interpretation:

Each coefficient β_j represents the expected change in response variable Y for a one-unit increase in predictor X_j , **holding all other predictor variables constant** ('ceteris paribus').

Source: Sridhar & Vijayalakshmi Ch. 5.3 (pp. 149-153)

Unit 3 8 Marks High Priority

Faculty docx Linear & ridge regression.docx Faculty PPT Ch. 5.4 NPTEL Week 2

Theory Link: Unit 3 § regularization

Explain the need for Regularization in regression. Differentiate between Ridge Regression (L2) and Lasso Regression (L1) with their loss functions, mathematical solutions, and feature selection properties.

Practice Problem:

Model Exam Answer

Solution:**Need For Regularization:**

When predictors are highly correlated (multicollinearity) or the number of features p is large relative to sample size n , OLS regression coefficients exhibit explosive variance, leading to extreme overfitting. Regularization introduces an explicit penalty on coefficient magnitude to shrink weights and enforce generalization.

Ridge Regression:

- **Loss Function (L2 Penalty):**

$$J_{\text{Ridge}}(\beta) = \frac{1}{2n} \|\mathbf{Y} - \mathbf{X}\beta\|_2^2 + \lambda \sum_{j=1}^p \beta_j^2$$

- **Closed-Form Solution:**

$$\beta_{\text{Ridge}} = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{Y}$$

- Adds $\lambda > 0$ along the diagonal, guaranteeing invertibility even when $\mathbf{X}^T \mathbf{X}$ is ill-conditioned.
- **Coefficient Behavior:** As $\lambda \rightarrow \infty$, coefficients shrink smoothly toward zero, but never reach exactly zero. Does not perform feature selection.

Lasso Regression:

- **Loss Function (L1 Penalty):**

$$J_{\text{Lasso}}(\beta) = \frac{1}{2n} \|\mathbf{Y} - \mathbf{X}\beta\|_2^2 + \lambda \sum_{j=1}^p |\beta_j|$$

- **Sparsity Property:** The sharp corners of the L1 diamond constraint boundary cause contours of the MSE loss ellipse to hit axes directly, forcing irrelevant feature coefficients to become **identically zero** ($w_j = 0$).

Unit 3 6 Marks High Priority

Ramaiah CIE-1 2026 Q2.c (3M)Faculty docx Logistic Regression.docxSEE May 2023 Q1 (6M)

[Theory Link: Unit 3 § logistic-regression](#)

Differentiate Logistic Regression and Linear Regression in machine learning. Why is Linear Regression unsuitable for classification problems?

Practice Problem:

Model Exam Answer

Solution:

Why Linear Fails Classification:

- **1. Unbounded Output:** Linear regression produces predictions in $(-\infty, +\infty)$, violating the mathematical constraint that probabilities must lie strictly in $[0, 1]$.
- **2. Sensitivity to Extreme Outliers:** An extreme positive point far to the right pulls the OLS regression slope down, inadvertently shifting the decision boundary and causing false negatives.
- **3. Non-Normal, Heteroscedastic Errors:** Binary classification residuals violate Gauss-Markov assumptions (residuals can only be $1 - \hat{y}$ or $-\hat{y}$, having variance that depends on $\hat{y}$).

Comparison Table:

Headers:

- Feature
- Linear Regression
- Logistic Regression

Rows:

- ['Target Variable Type', 'Continuous numerical variable y in $\mathbb{R}$.', 'Discrete categorical variable y in $[0, 1]$ ']
- ['Hypothesis Function', 'Linear: $\hat{y} =$

Unit 3 6 Marks Medium Priority

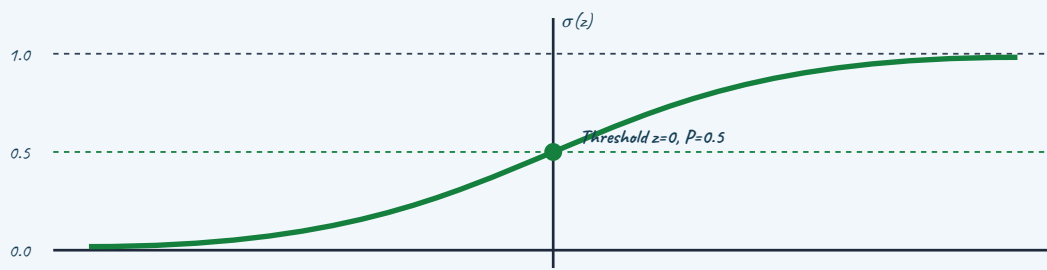
Faculty docx Logistic Regression.docxNPTEL Regression Week 3 Assignment

Theory Link: Unit 3 § logistic-regression

Explain the Sigmoid function, Odds Ratio, and Logit (Log-Odds) formulation in Logistic Regression. A logistic model produces linear score $z = 1.386$ for an input. Calculate: i. The odds, ii. The predicted probability $P(Y=1)$, iii. Final classification with threshold 0.5.

Practice Problem:

Model Exam Answer

Solution:Steps:

- i. Calculate Odds:

$$\text{Odds} = e^{1.386} \approx 3.9988 \approx 4.0$$

- ii. Calculate Predicted Probability $P(Y = 1)$:

$$P(Y = 1) = \frac{1}{1 + e^{-1.386}} = \frac{1}{1 + 0.25007} = \frac{1}{1.25007} = 0.79995$$

- iii. Classification Decision:

Since $P(Y = 1) = 0.80 > 0.50$ (decision threshold), the instance is classified into ****Class 1****.

Mathematical Foundations:

- Sigmoid Activation Function:

$$\sigma(z) = \frac{1}{1 + e^{-z}} = \frac{e^z}{1 + e^z}$$

Maps any input $z \in (-\infty, +\infty)$ to probability $p \in (0, 1)$. Derivative: $\sigma'(z) = \sigma(z)(1 - \sigma(z))$

Unit 3 6 Marks Medium Priority

Faculty PPT Chapter 5.3SEE July 2024 Q3

Theory Link: Unit 3 § polynomial-regression

What is Polynomial Regression? How is it implemented using linear regression techniques? Discuss the risks associated with selecting very high polynomial degrees.

Practice Problem:

Model Exam Answer

Solution:

Polynomial Regression is a form of regression analysis in which the relationship between the independent variable x and dependent variable y is modeled as an d -th degree polynomial:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \dots + \beta_d x^d + \epsilon$$

Implementation As Linear Model:

- Although the relationship between y and x is non-linear, the model is strictly **linear in its parameters β** .
- It is solved directly by defining transformed feature vectors: $z_1 = x, z_2 = x^2, \dots, z_d = x^d$.
- The transformed design matrix $\mathbf{X}_{\text{poly}}$ is fed directly into the standard OLS Normal Equation: $\beta = (\mathbf{X}_{\text{poly}}^T \mathbf{X}_{\text{poly}})^{-1} \mathbf{X}_{\text{poly}}^T \mathbf{Y}$.

High Degree Risks:

- **1. Severe Overfitting (Runge's Phenomenon):** High-degree polynomials oscillate wildly near the boundaries of the data interval, achieving near-zero training error but catastrophic test error.
- **2. Extreme Multicollinearity:** As powers x^k increase, columns of $\mathbf{X}_{\text{poly}}$ become nearly collinear, causing $(\mathbf{X}^T \mathbf{X})$ to become ill-conditioned and numerical solutions to become unstable.
- **Remedy:** Use Cross-Validation to select degree d (typically $d \leq 3$) and apply Ridge regularization.

Source: Sridhar & Vijayalakshmi Ch. 5.3 (pp. 154-158)

Unit 3 7 Marks Medium Priority

Faculty PPT Chapter 5.2NPTEL Week 1

[Theory Link: Unit 3 § gradient-descent](#)

Explain Gradient Descent optimization for Linear Regression. Compare Batch Gradient Descent, Stochastic Gradient Descent (SGD), and Mini-Batch Gradient Descent.

Practice Problem:

Model Exam Answer

Solution:*Gradient Descent Overview:*

Gradient Descent is a first-order iterative optimization algorithm to find the parameter vector $\mathbf{w}$ that minimizes cost function $J(\mathbf{w})$. Parameters are updated opposite to the gradient:

$$\mathbf{w} \leftarrow \mathbf{w} - \alpha \text{del}_{\mathbf{w}} J(\mathbf{w})$$

where α is the learning rate hyperparameter.

Mse Gradient Computation:

For MSE cost $J(\mathbf{w}) = \frac{1}{2n} \sum_{i=1}^n (\mathbf{w}^T \mathbf{x}_i - y_i)^2$, the partial derivative is:

$$\frac{\partial J}{\partial w_j} = \frac{1}{n} \sum_{i=1}^n (\mathbf{w}^T \mathbf{x}_i - y_i) x_{ij}$$

*Comparison Table:**Headers:*

- Algorithm
- Samples Per Update
- Convergence Behavior
- Computational Efficiency

Rows:

- ['Batch Gradient Descent', 'Entire training set (n samples).', 'Smooth, monotonic descent to global minimum.', 'Very slow on large datasets ($O(n$

Unit 3 7 Marks Medium Priority

SEE May 2023 Q1 (7M) Faculty PPT Chapter 5.4

Theory Link: Unit 3 § regularization

Define and compare the following regression evaluation metrics: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R^2). Why is Adjusted R^2 preferred in multiple regression?

Practice Problem:

Model Exam Answer

Solution:Definitions:

- **Mean Squared Error (MSE):** Average squared difference between actual and predicted values:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Penalizes large errors heavily due to squaring; units are squared.

- **Root Mean Squared Error (RMSE):** Square root of MSE:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

In same measurement units as target y ; highly sensitive to outliers.

- **Mean Absolute Error (MAE):** Average absolute difference:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Linear penalty, robust to extreme outliers.

- **Coefficient of Determination (R^2 Score):** Proportion of variance explained by model:

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}} = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

Out of CIE-1 Scope Questions (18 Questions)

ARCHIVED

The following questions were identified in past papers but fall outside the official CIE-1 syllabus. They are preserved here to ensure zero questions are dropped.

Paper / Origin	Question Summary	Exclusion Rationale
Ramaiah CIE-1 2026 (Portions L1-L13)	Consider dataset of 10 samples for predicting whether to Play Tennis. Compute information gain of attribute outlook using ID3 algorithm.	<u>Decision Tree Learning (ID3, Entropy, Information Gain) is out of CIE-1 scope. ML Unit 3 scope is strictly limited to Regression.</u>
Ramaiah CIE-2 2026 (Portions L15-L28)	Explain the working of the CART algorithm.	<u>CART Decision Tree Induction is out of CIE-1 scope.</u>
Ramaiah CIE-2 2026 (Portions L15-L28)	Apply AGNES Algorithm for distance matrix clustering.	<u>Hierarchical Clustering belongs to Unit 4, outside CIE-1 scope.</u>
Ramaiah CIE-2 2026 (Portions L15-L28)	Apply Naive Bayes classification algorithm for Outlook/Play dataset.	<u>Bayesian Learning belongs to Unit 4, outside CIE-1 scope.</u>
Ramaiah CIE-2 2026 (Portions L15-L28)	Explain the Elbow Method used for selecting the optimal value of k in clustering.	<u>K-Means Clustering belongs to Unit 4, outside CIE-1 scope.</u>
Ramaiah CIE-2 2026 (Portions L15-L28)	Age and Loan predictors in dataset. Apply K-NN Regression to find House Price Index (HPI) for Age=48, Loan=\$142,000 for K=3.	<u>Similarity-based learning (K-NN regression) is excluded from CIE-1 scope.</u>

Paper / Origin	Question Summary	Exclusion Rationale
Ramaiah CIE-2 2026 (Portions L15-L28)	Illustrate with an example DBSCAN clustering algorithm: core, border, noise points.	<u>Density-based clustering belongs to Unit 4, outside CIE-1 scope.</u>
Ramaiah CIE-2 2026 (Portions L15-L28)	Explain the working of Random Forest and AdaBoost algorithms with suitable examples.	<u>Ensemble Learning belongs to Unit 5, outside CIE-1 scope.</u>
SEE June/July 2026	For the Play-Tennis dataset, build the complete ID3 Decision Tree showing Entropy and Information Gain at each level.	<u>Decision Tree ID3 is out of CIE-1 scope.</u>
SEE June/July 2026	Naive Bayes classifier for Flu prediction; Linear SVM boundary and support vectors for bank fraud.	<u>Bayesian Learning and Support Vector Machines belong to Units 4 & 5.</u>
SEE June/July 2026	K-Means clustering algorithm, DBSCAN core/border/noise, Elbow method for k.	<u>Clustering belongs to Unit 4.</u>
SEE June/July 2026	DIANA hierarchical clustering, Single-linkage clustering, Expectation Maximization (EM).	<u>Clustering & EM belong to Unit 4.</u>
SEE June/July 2026	Bagging vs Boosting, Feature subset selection, Model selection using hypothesis tests.	<u>Ensemble learning belongs to Unit 5.</u>
SEE July 2024	ID3 Decision Tree with Gain Ratio; Naive Bayes with Laplacian correction.	<u>Decision trees and Bayes classifiers are out of CIE-1 scope.</u>
SEE July 2024	K-Means, Agglomerative Hierarchical, Perceptron, Backpropagation in Multilayer Neural Networks.	<u>Clustering and Neural Networks belong to Units 4 & 5.</u>

Paper / Origin	Question Summary	Exclusion Rationale
SEE May 2023	Decision Tree C4.5 split info; KNN classification with distance weights.	<u>Similarity-based learning and Decision trees are excluded from CIE-1 scope.</u>
SEE May 2023	Naive Bayes document classification, SVM dual formulation, K-Means clustering, Backpropagation.	<u>Bayesian, SVM, and Neural Networks belong to Units 4 & 5.</u>
MAKEUP 2024 & SUPPLE 2025	Ensemble AdaBoost, Random Forests, K-Means, Perceptron algorithm.	<u>Ensembles, Neural Networks, and Clustering belong to Units 4 & 5.</u>